Weekly Assignment 4 :
Dijkstra

September 2024

1. Apply Dijkstra's algorithm on the graph in Fig.1, starting at the node labelled 0. Specify the shortest distance for each node at each iteration.

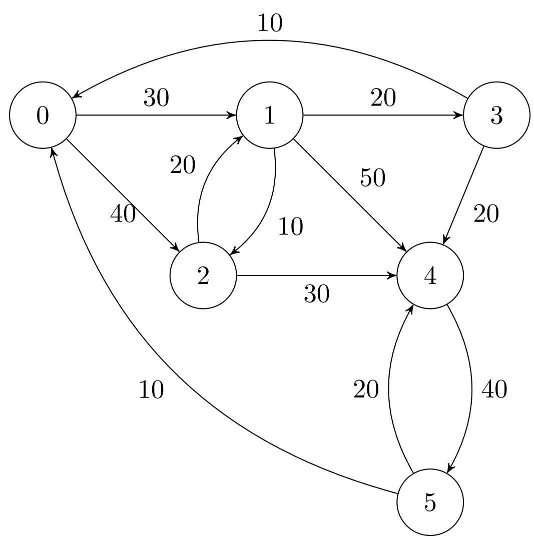

Figure 1: A weighted directed graph

2. Present an example that illustrates why Dijkstra's algorithm does not always work in case edges may have negative weights but there are no cycles of negative weight.

3. Airlines companies sell to travelers flights with or without connections, mostly depending on their airlines network. It can happen that secondary airlines are covered with small and slow aircrafts, hence a longer transit time. It can then be faster to fly via another city (connecting flight) rather than taking a direct flight, but that's not always the case. As connections are unpleasant, direct flights are preferred if the total flight time is equal. Otherwise, if the direct flight takes longer, then the connecting flight is preferred.

   Let us define $easiest[destination]$ = minimum number of successive flights in a shortest path from $origin$ to destination.

Figure 3: Airlines network.

In our airlines example, the *easiest* values are (for origin Madrid):

- 0 for Madrid.
- 1 for Amsterdam, Berlin, Rome.
- 2 for Stockholm, Kyiv.
- 3 for Moscow.

We can transform the problem of the airline company into a graph problem. Give an efficient algorithm taking as input a weighted graph $G = (V, E)$ and a source vertex $s \in V$, and as output the values of *easiest* for all vertices. Discuss correctness and efficiency of your algorithm.

4. Let $G = (V, E)$ be a weighted, directed graph with nonnegative weight function $w\colon E \to \{0, 1, \ldots, W\}$ for some non-negative integer W. Modify Dijkstra's algorithm to compute the shortest paths from a given source vertex s in $\mathcal{O}(W|V| + |E|)$-time. Discuss the correctness and complexity of the modified algorithm. **Hint:** Replace the min-heap priority queue with a simpler data structure.

5. Given the **max**-heap below:

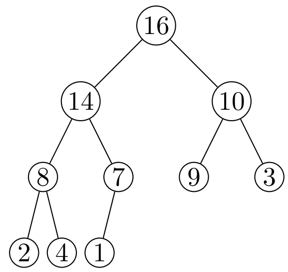

Figure 4: A Max-Heap with 10 elements

(a) Insert 11, show all intermediate heaps.

(b) Remove 14, show all intermediate heaps.